

CRVideo: Cross-reality 360° Video Social Systems Exploration

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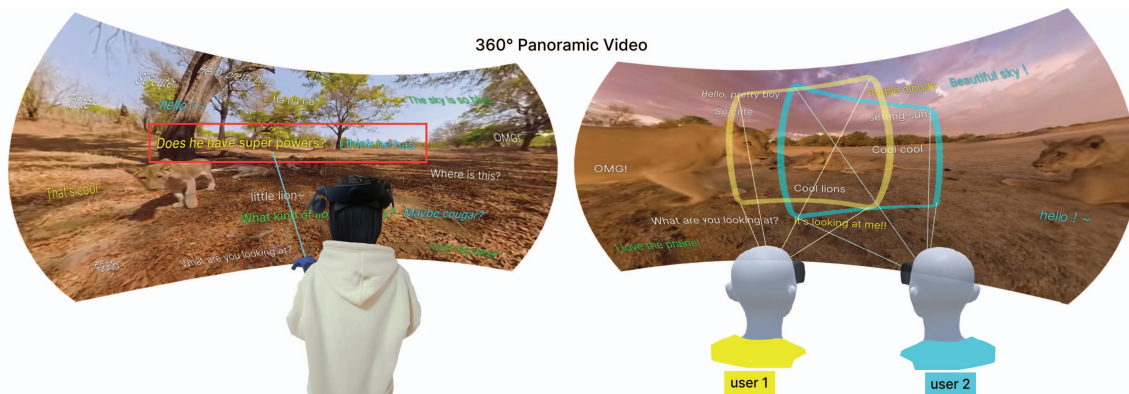


Figure 1: Illustration of 360° video social system: Left: the user watches bullet screen and selects online bullet comments to reply. Right: two users establish a collaborative relationship through bullet comments, jointly watch the 360° video and interact with each other.

ABSTRACT

Cross-reality systems provide visual representations at different levels of virtuality/physicality and offer smooth transitions between them to users. These systems also support multiple modes of interaction, and coordinate collaboration between users with multiple realities. In this paper, we explore a cross-reality 360° video social system, aiming to enable users with different personalities to get a social experience consistent with their psychological changes and behavioral habits when watching 360° video. We designed the prototype system CRVideo, conducted a preliminary user survey and a user study to research the social requirements of users and verify the effectiveness of our cross-reality system. The results show that the social habits of extroverted and introverted users in video platforms differ greatly, and the system across realistic levels can meet the psychological change curve of users in different periods.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies

1 INTRODUCTION

360° video has appeared on major video websites at home and abroad, such as YouTube, Niconico, and Bilibili. With the advancement of commercial omnidirectional cameras and head-mounted displays (HMDs) and video processing technology, users can rotate their heads in the virtual reality (VR) environment to experience video content in a more natural and immersive way [9]. However, VR devices completely separate the virtual and real environment, which makes a single user's head-mounted 360° video experience

lack a sense of social presence and can easily lose focus, thus often causing fatigue and confusion.

To solve the above problems, there are two existing solutions: (1) Enhance the sociability of 360° videos by simulating people's communication habits (gaze, gesture, voice, etc.) in physical reality and providing collaborative viewing and interaction experience. (2) Strengthen the understanding of video content and promote the interaction between viewers through bullet comments, so that video viewing is no longer limited to passively receiving inherent information. The two methods have different focuses and complementary advantages. Collaborative viewing is a synchronous communication and the migration of physical reality, while bullet comments interaction is an asynchronous communication and the construction of virtual internet. Besides, the user's mental model is greatly affected by personality [5]. For introverted users, they may prefer to receive bullet comments or occasionally express opinions in the virtual world, while for extroverted users, they may be more accustomed to communicating with others naturally and directly in an environment close to the reality. At the same time, the psychological curve of the same user is not static, and the user's behavior will be affected by various external and internal factors and change at random, which makes it difficult for a single collaborative viewing or bullet comments interaction to meet the needs of all users at any time.

We surveyed existing video sites and found that the existing video platforms only focus on one aspect of collaborative viewing or bullet comments interaction, and lack of compatibility for different personality groups and the conversion between multiple interaction modes. From the two aspects of user personality and user psychology, based on the shortage of current video platform functions, we have the motivation to study whether users need this innovative platform, and then verify whether this video platform is meaningful through examples. In this paper, we innovatively explore a cross-reality (CR) social platform to meet the real-time virtual/real interaction preferences of different user groups while watching 360° videos. Cross-reality systems provide users with different weights of virtual or physical reality, and smooth transitions in the Reality-Virtuality (RV) continuum [8], which makes the communication and collabora-

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tion between users in different virtual/physical levels more flexible and diverse. Based on the characteristics and design principles of cross-reality systems, we developed a prototype system CRVideo to support two social modes of bullet comments interaction and collaborative viewing in 360° video space, and provide seamless conversion between the two to meet the possible social desire of different user groups (see Fig. 1).

We conducted a preliminary user survey and a user study to verify the significance of the cross-reality system design and the effectiveness of our prototype system. Through simple quantitative analysis and qualitative evaluation, we know that CRVideo can not only adapt to the social behavior preferences of users with different personalities, but also meet the social psychological curve of the same user in different states.

2 RELATED WORK

In this section, we discuss the related research in 360° video experience and cross-reality system respectively. In order to enhance the sociability of 360° videos, researchers have made efforts from two aspects: collaborative viewing and bullet comments interaction, and our work combines the respective advantages of these two interaction modes. In addition, through the study of cross-reality systems, we propose our system that can transition and collaborate between real collaboration and virtual bullet comments.

2.1 360° Video Experience

360° video allows viewers to choose the target view in all directions according to their interests [14], but when the video content is complex, it will be confusing and exhausting. Collaboration in VR can not only help supplement other content that users overlook, but also enhance the sense of presence and engagement through more natural and effective interaction behaviors. Nguyen et al. [11] proposed CollaVR system to support collaborative video viewing and communication in a VR head-mounted environment. However, synchronous cooperative interactions cannot be generalized to online streaming platforms, face bandwidth challenges in processing large amounts of real-time data, and require all users to be online simultaneously. The asynchronous communication mode of bullet comments solves the limitation of time and space and enhances social interaction. Lee et al. [4] used bullet comments in online educational videos to significantly enhance student engagement in learning. Li et al. [6] compared four display methods and two insertion methods of bullet comments in 360° videos, analyzed and summarized the most popular display and interaction mechanisms among users. Our CRVideo system complements each other's strengths and is designed to provide users with diverse and flexible 360° video interaction methods.

2.2 Cross-reality System

The RV continuum is a classical paradigm for classifying extended reality applications [8]. However, the positioning of an application on the continuum may change throughout the experience, and the user's psychology and behavior may not match the initial positioning of the application. George et al. [1] explored seamless, bidirectional transitions on the RV continuum. Jorge et al. [3] proposed a cross-reality platform called Anatomy Studio, which allows remote interaction between students and teachers to exchange opinions and correct errors instantly while performing remote anatomy training. Yu et al. [13] developed guqin systems VirGuqin and MR-Guqin, which use VR and MR technology respectively, to provide targeted immersive experiences for two groups: users who have not been exposed to guqin and guqin beginners. Gottsacker [2] discusses the balance between virtual and physical by studying cross-reality interactions between VR users and non-VR interrupters. Pagador et al. [12] proposed HiruXR system, which aims to design adaptive visualization and interaction interfaces across 2D and XR devices and to support collaboration across platforms. Similar to these works, we

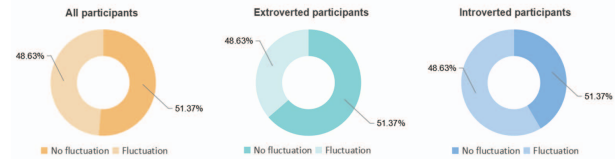


Figure 2: The proportion of participants with different personalities who indicated that they experienced psychological fluctuations.

envision a cross-reality system that allows user groups with different personalities to choose preferred video social behaviors according to actual requirements, and provides collaboration and interaction across different levels of virtuality/physicality.

3 CRVIDEO

Based on the work foundation of enhancing sociability in 360° video and the innovation of cross-reality technology, we combine the respective advantages of virtual bullet comments and natural reality communication, propose a 360° video social platform that across realistic levels, and explore its significance and value for users. We first conduct a preliminary user survey to collect personality information, social preferences, and past psychological experiences from a large number of users. Through user survey, we analyze the different requirements of video viewers for cross-reality 360° video social systems, and select the experimental subjects from them. We designed and implemented the interaction procedure of CRVideo, allowing users to experience bullet comments interaction and synchronous collaboration in 360° video space. We collected each user's behavioral data during the experiment, evaluated and verified the effectiveness of CRVideo through quantitative analysis, questionnaire survey and semi-structured interview after the experiment.

3.1 Preliminary User Survey

We recruited 278 users (127 male, 151 female) from video websites to participate in this survey, aging from 17 to 32 (Mean = 23.11, SD = 2.74). All users watch online videos regularly. We asked them to subjectively define their personality as extroverted or introverted. In addition, we obtained the implicit personality of each participant through the 'Extroversion - Introversion' preference set of the Myers Briggs Type Indicator (MBTI) [10]. Matching the participants' subjective definitions to their implicit performance, of all the participants, 113 belonged to the extroverted group, 142 belonged to the introverted group, and 23 were unclear about their personality or did not have a self-definition that matched the MBTI results. We asked each participant to answer several questions we designed, such as, "Are you an extrovert or an introvert?" "What kind of interaction do you prefer when watching videos?" "Do your preferences never change or do they change often?" "What limitations have you faced when socializing on video platforms?" All 113 extroverted participants showed a strong desire to share and discuss during the video-watching activity, and 41 of them also indicated that they often needed alone time. Among 142 introverted participants, 134 said they often spent a lot of time to learn other viewers' opinions on videos by watching bullet screens, while the other 8 participants had no habit of paying attention to bullet screens. Meanwhile, 83 of the 142 participants had the idea of sharing or co-watching a video while watching it, accounting for 58.5% of the introverted group. At the same time, most participants reported that it was difficult for one video platform to satisfy all their social preferences, and they sometimes jumped between multiple video apps, which was cumbersome.

Through the survey results, we found that the social habits of extroverted and introverted groups were quite different when watching

videos, and 48.6% of the participants showed fluctuations in psychological curves and changes in behavioral preferences, especially in the introverted group, which indicated that the introverted group was more likely to have fluctuations in social psychology (see Fig. 2).

3.2 Participants and Apparatus

According to the results of the preliminary user survey, we selected 24 users (11 males and 13 females), including 12 in the extroverted group and 12 in the introverted group, aging from 19 to 28 (Mean = 22.58, SD = 2.29). All participants had normal vision or corrected vision, and 5 of them did not have any prior VR experience. All participants had seen videos with bullet comments, but 9 had not been exposed to 360° videos.

The experiment was conducted in a laboratory with a physical tracking space of 3m×3m. An HTC Vive Pro Eye was used as the HMD with a refresh rate of 90Hz. Use the HMD's own audio, and use the matching handle as the hand operation terminal. For the development environment, the Unity3D engine was selected, running on AMD Ryzen™ 5 3600 CPU with Nvidia Geforce GTX1070 GPU and 32GB RAM.

3.3 Interaction Design

We collected four 360° videos with bullet comments from YouTube (one for training period and three for formal experiment), and clipped the videos to 2 to 3 minutes. Similar to previous work [6], we adapt the presentation form of bullet comments to a horizontally sliding spherical display. Manually adjust the time and space position of the bullet comments according to the timestamp and content of the bullet screen data in the downloaded video resources, so as to ensure the bullet screen content is a comment or description of the specified view in a certain time. CRVideo takes bullet screen as the initial social state, and each user can watch 360° videos and bullet comments through VR devices. The ray emitted by the handle is used to specify a certain position in the field of vision, and the content of the bullet comments can be input by voice. Using speech recognition technology to convert speech information into text and float it on the specified time and space position. The status information of senders is recorded, and the comment sent by online users is highlighted, so that other users can see which bullet comment owners can collaborate with them in real time.

As for how users perceive the presence of other partners, we were greatly inspired by Lin et al.'s spatial picture-in-picture previews [7], on which we designed our floating thumbnail view to prompt and guide the replied user to rotate the perspective, so as to establish a collaborative relationship (see Figure 3). When a user replies to a certain bullet comment, the area where the reply occurs will be captured to form a floating thumbnail view, and then map the three-dimensional space vector of this area relative to the user 1 into the two-dimensional plane vector of the thumbnail view relative to the user's current field of view (see Figure 4). This way can intuitively guide the rotation of user 1 in the horizontal and vertical directions. If user 1 chooses to return to the field where he/she has been replied, when the overlapping view of user 1 and user 2 is detected, it is regarded as the establishment of collaborative relationship, and the timeline of both parties is pulled to the time point when the reply occurs. Collaborative viewing users can visualize each other's visual field to track what the other is focusing on, and share opinions through multimodal interactions such as voice conversations and ray-pointing. At the end of the communication, users can exit the collaboration state with the handle button or speech command and return to the original bullet comments interactive interface.

3.4 Experiment Procedure

We first asked and recorded the basic information of participants, including serial number, age, gender, vision, VR and 360° video experience, etc. Participants were informed of the experimental



Figure 3: Illustration of prompt and guide mechanisms: when user 1 is replied by user 2, a floating thumbnail view pops up.

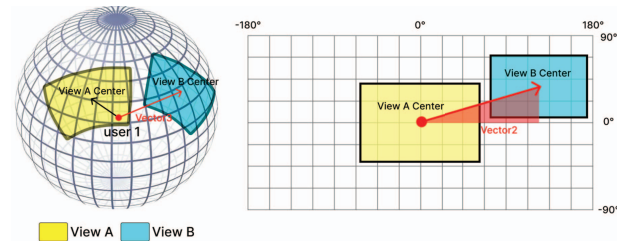


Figure 4: Mapping of three-dimensional space vector to two-dimensional plane vector.

objectives and operational procedures, read and signed the experimental consent form. After users wore the HMD, the training video was used to familiarize the user with multimodal interaction in 360° video space.

In the formal experiment, each user watched the remaining three formal 360° videos and freely chose social behaviors according to the interaction mode learned during training period. The videos were viewed in random order and there was a 5-minutes break between two videos. We collected six behavioral data, including the number of times of sending bullet comments (reply comments and non-reply comments), the number of times of establishing collaboration (active collaboration and passive collaboration), and the duration of collaborative viewing (active collaboration and passive collaboration), when extroverted and introverted participants watched 360° videos. Finally, participants were asked to fill out a questionnaire and conduct a semi-structured interview to collect free subjective feedback. Each participant spent an average of 50 minutes and eventually received a gift.

4 DISCUSSION

In total, we collected 432 user behavioral data (24 participants×3 360° videos×6 quantitative metrics). Only 1 of the 12 extroverted participants never sent a bullet comment, 8 of the remaining 11 responded to online bullet comments, and 3 only sent bullet comments in response to the video content. A total of 10 extroverted users participated in collaboration, of which 7 initiated collaboration actively and 8 accepted collaboration passively. Four of the 12 introverted participants did not send bullet comments, 5 of the remaining 8 responded to online bullet comments, and 3 sent non-reply comments. A total of 6 introverted users participated in collaboration, of which 3 initiated collaboration actively and 4 accepted collaboration passively. Fig. 5 shows the average number of times that 24 users sent bullet comments, established collaboration, and the average duration of collaboration in three 360° videos. Through the analysis, we can learn that each social metric of the extroverted group is higher than the introverted group, and there are significant differences between extroverted group and introverted group except for passive cooper-

Table 1: Behavioral data statistics in user study. Significant differences are indicated with * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Behavior metrics	Personality	Mean	SD	F	p
Send bullet comments (reply / non-reply)	Extroverted	11.5 (7.333 / 4.833)	6.038 (5.821 / 2.623)	7.799 (6.098 / 3.273)	0.011** (0.022** / 0.084**)
	Introverted	5.333 (2.5 / 2.833)	6.697 (3.477 / 2.791)		
Establish collaboration (active / passive)	Extroverted	4.417 (2.5 / 1.917)	2.875 (2.505 / 1.564)	5.291 (4.435 / 2.091)	0.031** (0.047** / 0.162)
	Introverted	2.083 (0.833 / 1.083)	2.021 (1.115 / 1.24)		
Duration of collaboration (active / passive)	Extroverted	5.85 (2.873 / 3.067)	3.734 (2.827 / 2.66)	8.499 (4.748 / 4.644)	0.008*** (0.040** / 0.042**)
	Introverted	2.05 (0.875 / 1.175)	2.539 (1.1 / 1.474)		

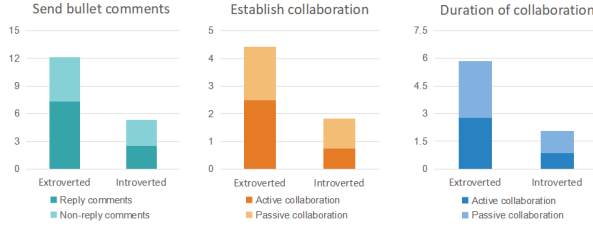


Figure 5: Comparison between extroverted and introverted users regarding the number of bullet comments sent, the number of collaboration established, and the duration of collaboration.

ation (see Table 1). Users in the extroverted group also showed a willingness to avoid social interaction, even without sending one bullet comment or engaging in collaboration throughout the whole experiment. Similarly, several users in the introverted group showed a tendency to actively communicate with other viewers. Compared with the extroverted group, introverted group is more likely to have fluctuations in psychological and behavioral preferences, which is also in line with the user requirements we have investigated.

At the end of the experiment, we designed a 7-Likert scale questionnaire with four questions and asked participants to give their subjective scores (The score ranges from 1 to 7, 1 stands for very dissatisfied and 7 stands for very satisfied). The four questions and scores were: (1) The immersion level of the 360° video bullet comments space (Mean = 5.17, SD = 0.99); (2) The degree of natural interaction in collaborative viewing mode (Mean = 5.08, SD = 1.04); (3) The smoothness of the transition between the two reality levels (Mean = 5.33, SD = 1.31); (4) Preference for cross-reality video social system (Mean = 5.92, SD = 0.95).

In semi-structured interviews, we ask some questions and encourage users to freely express their feelings. Our questions included, but were not limited to, “Does CRVideo meet your video social needs?” “What are you pleased with about it?” “What bothered you or discouraged you from using CRVideo?” All 21 users expressed their preference for our CRVideo prototype system, while the other 3 users were annoyed by the need to drag timeline frequently, visual interruptions, and other reasons. More than half of the users explained that the cross-reality video social system allows them to switch social behaviors at will, choosing the current most preferred interaction mode. We explained the behavior of the other group for each of the two groups, all users reported that the other group was completely different from most of their states. Most users indicated that their emotions and psychology changed from time to time, even though they were very confirm their personality and social habits. In summary, the good scores of questionnaire and the subjective evaluation obtained from semi-structured interview further verify the bright prospect about the application of cross-reality technology in 360° video social system.

5 CONCLUSION AND FUTURE WORK

In this paper, through user survey and analysis, we find that extroverted and introverted users have very different social habits when they watch videos and interact with other viewers. Moreover, due to psychological changes and other factors, the same user has different requirements for video platforms in different states. Combining the characteristics of cross-reality systems and the diversity of user requirements, we provide a prototype system CRVideo for users with both virtual bullet comments interaction and realistic collaboration space, and achieve a smooth transition between these two environments with different degrees of virtuality/physicality. Through the user study, we can verify that the cross-reality 360° video social system is an effective exploration to meet the different psychological and social behavior habits of users.

Our current research is in the preliminary exploratory stage. Since 360° videos are still mostly viewed on 2D screens today, we will consider adaptive display and interactive collaboration across devices in the next stage of our research. In addition, our user evaluation metrics are not clear enough in the definition and identification of psychological fluctuations. In our future work, eye tracking technology will be used to measure user behavior and analyze the user psychological curve more deeply by capturing visual gaze data, so as to support our research conclusions. Finally, we sincerely hope that our research results will not only be limited to 360° video space, but also provide some useful inspiration for designers and researchers of other cross-reality systems.

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